

FOSSEE

eSim Semester Long Internship — Autumn 2026

TASK 7

PCB DESIGN REPORT

SpectraSense

Two-Layer LM741-Based Signal-Conditioning PCB

Field	Details
Candidate Name	Sinjini Biswas
eSim Project	SpectraSensesub
Deliverable	Task 7 — PCB Design Report
Design Type	Two-layer, through-hole analog PCB
Primary Active Device	LM741 operational amplifier

Technical record of schematic capture, footprint assignment, PCB layout, routing, DRC and 3D verification.

1. Project Overview

2. Circuit Description

3. Schematic Design

4. Footprint Assignment

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1. Project Overview

SpectraSense is a compact, two-layer, through-hole analog PCB implementing a single-stage LM741 operational-amplifier signal-conditioning circuit. The design accepts an external sensor signal through a dedicated input interface, processes it through an inverting op-amp stage with resistive feedback, and provides a separate power-entry interface with local decoupling.

The implementation follows the PCB workflow specified for Task 7: schematic capture, footprint assignment, Sch to PCB Update, component placement, routing, Design Rule Check (DRC), and 3D verification. The internship task requires at least two copper layers and evaluates schematic organization, footprint mapping, layout/routing practice and verification.

The board is organized around four functional groups: the external sensor interface, the LM741 analog processing stage and feedback network, an auxiliary diode branch, and the power-input/decoupling section.

2. Circuit Description

Sensor Input Interface — J1 (SENSOR_IN)

J1 is a two-pin through-hole header forming the signal-level interface. Pin 1 carries the sensor signal into the LM741 inverting-input node, while pin 2 provides the common ground reference.

LM741 Analog Processing Stage — X1

X1 is an LM741 configured as an inverting amplifier. The inverting input (pin 2) forms the summing node shared with the sensor input and feedback resistor. The non-inverting input (pin 3) is tied to ground. The output is taken from pin 6 and returned through R1. The unused offset-null pins (1 and 5) and NC pin (8) remain unconnected.

Feedback Network — R1 (100 k Ω)

R1 closes the feedback loop from the LM741 output to the inverting-input/sensor summing node, defining the stage's inverting-gain relationship.

Auxiliary Diode Branch — D1 (1N4148)

D1 is a separate branch rather than a series element in the sensor signal path. One terminal connects to the simulation-only current source i1, while the other terminates on a node with no further connection. It is therefore not part of the main SENSOR_IN–R1–X1 signal chain.

Power Input and Decoupling — J2 and C1

J2 is the power-entry header. Pin 1 feeds the POWER_IN/decoupling node and pin 2 returns to GND. C1 (100 nF) is connected across this pair to provide local bypass decoupling. The current schematic uses simulation-only V1 for the LM741 V+ pin; V1 is not a physical PCB component.

The SPICE-only sources V1 and i1 exist for simulation and are intentionally absent from the populated footprint set.

3. Schematic Design

The schematic captures the physical components and the two simulation-only sources used for ngspice analysis. The main analog path is centered on the LM741, with R1 providing feedback and J1 providing the external sensor input.

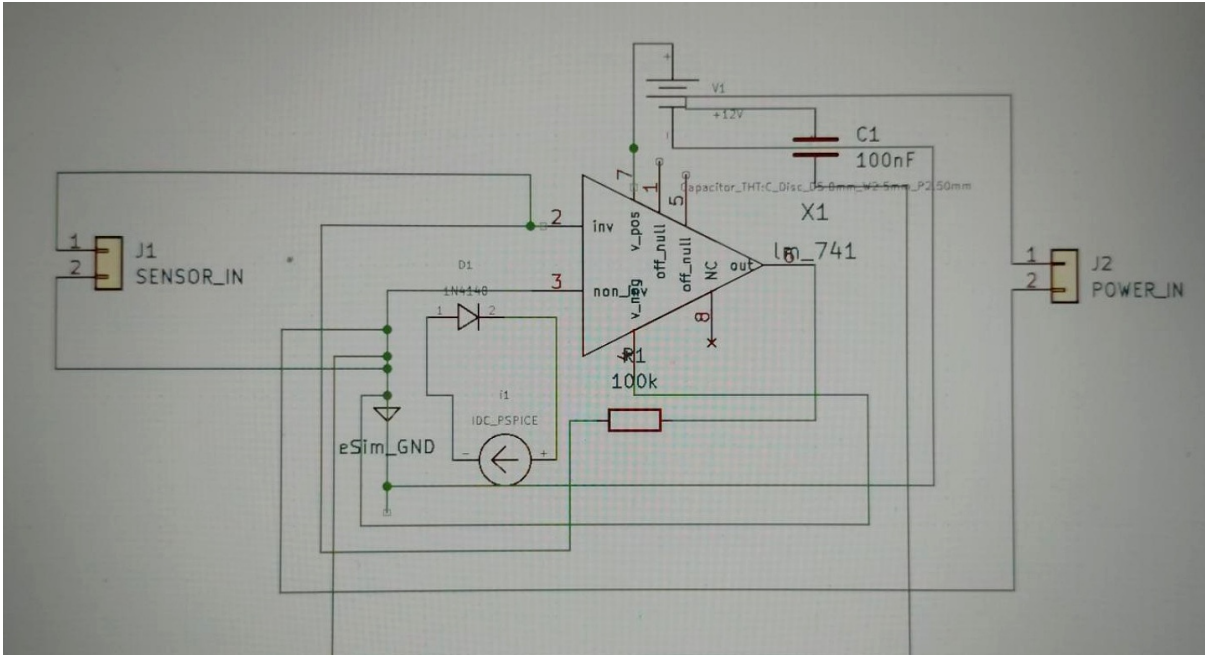


Figure 1. Enhanced eSim schematic capture of the SpectraSense circuit.

The supplied capture is retained as project evidence so that the report reflects the actual work product rather than a reconstructed drawing.

4. Footprint Assignment

Each physical component is assigned a footprint appropriate to its package and through-hole construction. The six populated parts are transferred to the PCB through Sch to PCB Update.

Ref.	Component	Value	Assigned Footprint	Package / Type
X1	LM741	Op-amp	Package_DIP:DIP-8_W7.62mm	8-pin DIP, THT
R1	Resistor	100 kΩ	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	Axial THT
D1	Diode	1N4148	Diode_THT:D_DO-35_SOD27_P10.16mm_Horizontal	DO-35 axial THT
C1	Capacitor	100 nF	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P2.50mm	Disc ceramic THT
J1	Pin header	SENSOR_IN	Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical	1×2, 2.54 mm
J2	Pin header	POWER_IN	Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical	1×2, 2.54 mm

The footprint choices keep the design consistent with a hand-assemblable prototype: DIP-8 for the LM741, standard axial footprints for R1 and D1, a compact disc footprint for C1, and standard 2.54 mm headers for the external interfaces.

5. PCB Layout

The board uses two copper layers, F.Cu and B.Cu, within a rectangular 45.72 mm × 25.40 mm (1.8 in × 1.0 in) outline. Placement follows the functional structure of the circuit.

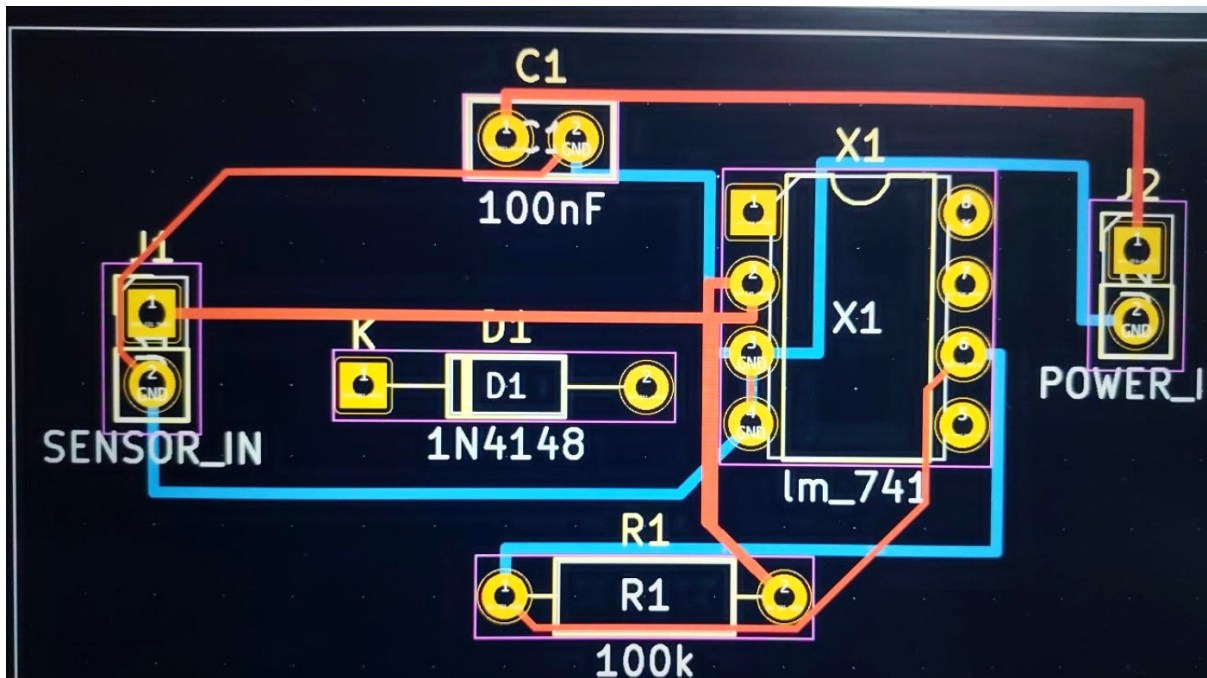


Figure 2. Enhanced routed PCB top-view capture using the existing PCB layout.

J1 and J2 are positioned near opposite board edges for accessible external connections. X1 is centrally placed, R1 is kept close to the feedback region, C1 is kept near the power-entry area, and D1 occupies the separate auxiliary branch region.

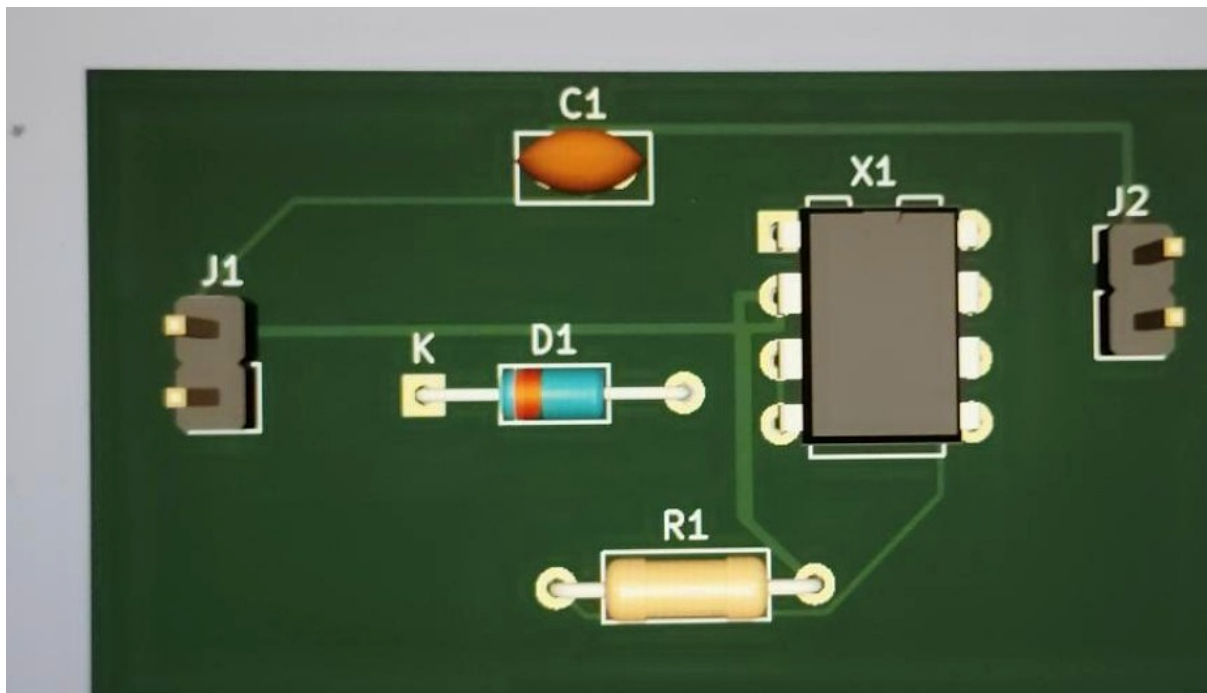


Figure 3. Enhanced 3D Viewer top/orthographic capture of the populated SpectraSense PCB.

6. Routing Methodology

Routing was performed after schematic-to-PCB transfer. The completed board uses both F.Cu and B.Cu and does not require dedicated vias. Since all six physical components are through-hole, their plated pads provide natural electrical continuity between the two copper layers.

The SENSOR_IN/summing node is routed directly around the LM741 and feedback network. GND and the feedback net use both copper layers where that provides a clearer path. The J2-to-C1 power/decoupling connection is kept short.

Routing Aspect	Implementation
Copper layers	F.Cu + B.Cu
Board construction	Two-layer through-hole
Dedicated vias	0
Nominal track width	0.25 mm; some routes 0.4 mm
Nominal clearance	0.20 mm
Signal routing	Direct SENSOR_IN/summing-node routing
Power routing	Short J2-to-C1 connection
Unused LM741 pins	Pins 1, 5 and 8 left unrouted

The routing is deliberately conservative and readable for a compact prototype PCB while satisfying the two-layer requirement.

7. DRC Verification

The supplied final DRC result reports zero violations and zero unconnected items, with zero errors and zero warnings.

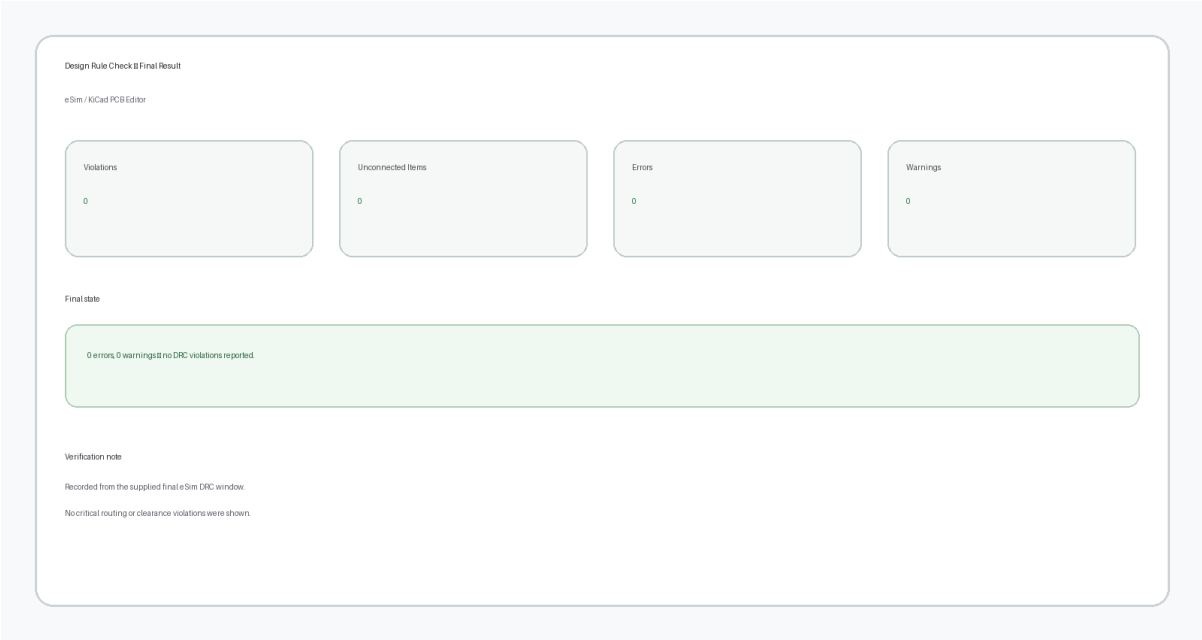


Figure 4. Final DRC result recorded from the supplied eSim DRC window.

DRC Item	Final Result
Violations	0
Unconnected items	0
Errors	0
Warnings	0
Final state	0 errors, 0 warnings

No critical routing, clearance or footprint-rule issue is indicated by the supplied final result.

8. 3D PCB Visualization

The 3D Viewer provides a physical visualization of the populated board and is used to check component presence, package geometry, orientation and the overall board outline.

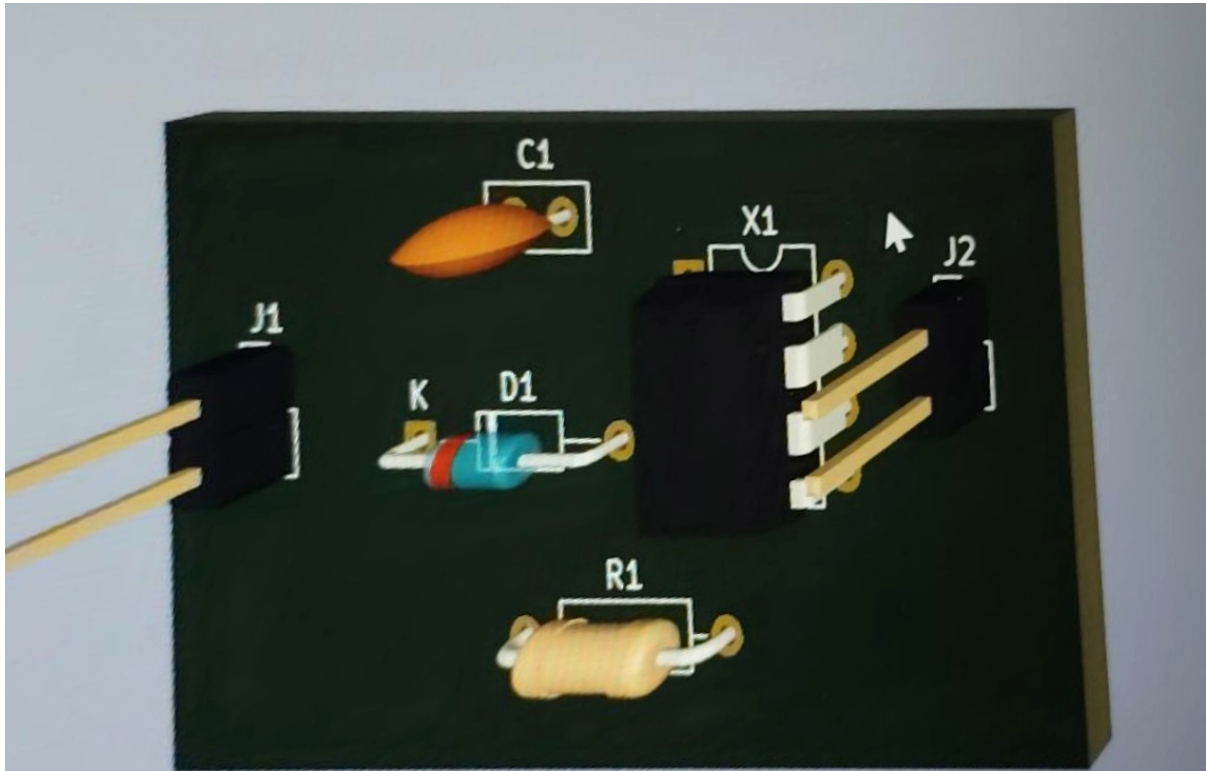


Figure 5. Enhanced 3D Viewer perspective capture of the populated SpectraSense PCB.

The 3D view shows the six physical footprints: the DIP-8 LM741, axial resistor R1, DO-35 diode D1, disc capacitor C1, and the J1/J2 pin headers. The board remains within the rectangular Edge.Cuts outline.

The 3D representation complements the 2D layout by providing a physical sanity check of the selected through-hole packages and their placement.

9. Design Considerations

Two-layer architecture. A two-copper-layer stack is appropriate for the net count and pin density of this circuit and satisfies the task's minimum layer requirement.

Through-hole construction. All populated components use through-hole footprints, keeping the board straightforward to hand-assemble and suitable for a laboratory/prototyping context.

Footprint suitability. Each footprint matches the physical package implied by its component: DIP-8 for the 741, DO-35 for the diode, axial THT for R1, disc THT for C1, and 2.54 mm headers for J1/J2.

Placement. Connectors are placed toward the board perimeter, the LM741 is central, and passives are kept close to the nodes they serve.

Routing. Two-layer routing is used to keep connections direct while maintaining readable geometry. Through-hole pads provide natural layer-transition points without requiring dedicated vias.

Power decoupling. C1 is placed directly across the POWER_IN and GND nodes to provide local bypassing at the power-entry point.

Manufacturability. Track widths in the 0.25–0.4 mm range and the 0.2 mm clearance are practical for this compact through-hole board.

Schematic-level limitations retained as-is

D1's pin 1 terminates on an isolated node and is not part of the main sensor/feedback path. The LM741 V+ pin is driven by the simulation-only V1 source rather than by an on-board POWER_IN net. These are properties of the supplied schematic connectivity and are not treated as routing defects introduced during PCB implementation.

10. Conclusion

SpectraSense demonstrates the complete PCB design workflow required by Task 7: schematic capture, appropriate through-hole footprint assignment, schematic-to-PCB transfer, organized component placement, two-layer copper routing, DRC verification and 3D visualization.

The final board is compact, functionally organized and suitable for a prototype-style implementation. The supplied DRC result records zero violations, zero unconnected items, zero errors and zero warnings, providing the final rule-check evidence for the routed design.

Overall, the implementation emphasizes a deliberate physical realization of the circuit: the sensor and power interfaces remain accessible, the amplifier and feedback network are centrally organized, the decoupling component is near the power entry, and the two-layer routing is direct and readable.

Submission Evidence Summary

Required Stage	Evidence Included
Schematic capture	Figure 1 — supplied eSim schematic capture
Footprint assignment	Section 4 — complete footprint mapping table
PCB layout / placement	Figure 2 and Figure 3
Routing methodology	Section 6 — layer, width, clearance and routing summary
DRC verification	Figure 4 — final result; 0 errors / 0 warnings
3D Viewer	Figure 5 — supplied populated-board 3D view